PROJECT

WOOF MATE

A smart pet toy designed to go beyond simple automation—providing comfort, engagement, and emotional companionship when owners are away.

- *Creative Technology*
- *Pet Toy Design*
- *User Experience*

GREEN COMBO

A board game designed to help people understand sustainable stores in China and build sustainable living habits.

#Service Design

#Social Innovation

#Board Game Design

PETIFIED MARKET

A specially made checkout device, it satirizes people who, while resisting consumerism, unknowingly fall into a new consumerist trap.

#User Experience

#Installation design

#brand design

LIFE MANCER

A critical speculative design that satirizes parents' growing obsession with control and the narrow, standardized definition of "success."

#Critical Speculative design

#Wearable devices

MEMOJI

An AI-powered family communication service app for seniors. It helps seniors rebuild a communication experience of "being seen, understood, and responded to" in the digital age.

#Digital Applications

#Media Communication

#Inclusive Design

WOOF MATE

Intelligent Pet Toys for Absent Owners

In fast-paced urban life, more and more pets (dogs and cats) are left alone at home for long hours. Beyond physical needs like feeding, they face emotional stress, loneliness, and boredom. This project explores how a smart pet toys can go beyond automation—becoming a subtle presence that comforts, engages, and connects when owners are away.

Background

Lonely dog in House

8~12H

In modern urban life, **60% dogs are left home alone for 8 ~12 hours** while their owners are at work.

Smart but inhuman

Due to the fast-paced life, the owner don't have time and energy to compare their dogs.**They add more toys, leisure, smart devices make the dogs comfortable.**

However, these product oversee the pets emotion need and ignore **the pet is still alone in the house.**

Research

What makes the device inhuman?

1 Lack of Emotional Feedback

Most products can't interpret or respond to emotional cues like **loneliness, excitement, or distress—making the interaction feel robotic.**

2 One-Way Interaction

Devices often act on the pet (e.g., dispense food) **without reacting to the pet's behavior, mood, or presence.**

3 Generic Functionality

Many devices follow rigid schedules or pre-set functions, ignoring the dynamic, personalized nature of real relationships.

4 Ignoring Pet Agency

Most systems don't let the pet initiate interaction—they're passive receivers, not active participants in the relationship.

Persona

Zoe

The Caring but Busy Owner

Age: 30
Occupation: UI Designer
Location: Shanghai
Pet: Corgi, male, 2 years old
Tech Literacy: High (comfortable with apps, smart home devices)

Goals:

1.Keep his dog **emotionally comforted** during long work hours.
2.Maintain **a sense of presence at home** while away.
3.**Feel connected to her pet,** not just monitor it.

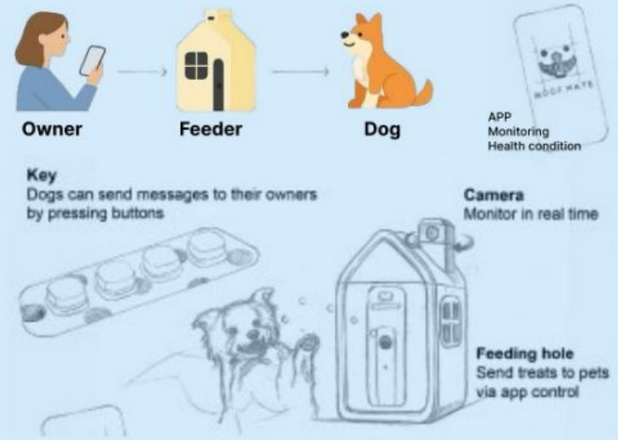
Frustrations:

1.Existing devices **feel cold and robotic**
2.**No feedback from pet,** making remote interaction feel pointless
3.**Guilt over leaving her dog alone,** especially on overtime days

Insight	Design Direction
Pets need emotional companionship not just functional feeding	Introduce voice playback, soothing elements, and emotion-responsive feeding
Interaction should be two-way rather than one-directional control	Allow pets to initiate interactions the device responds automatically
Owners seek emotional connection, not just monitoring	Provide emotional status visualization and features that simulate the owner's presence

Initial Concept

Based on my insight, I carried out the initial design: Owners can remotely interact with their pets at home through the feeding device.



Interview

I shared my design ideas and prototype with a professional veterinarian.

Amy, veterinarian

From a clinical and behavioral perspective, dogs are naturally motivated by physical stimuli—movement, tactile feedback, or scent—rather than abstract cues like virtual sounds.

Your idea is good, but it might be necessary to consider it more from the perspective of the dog's physiological habits. Your device attracts them through sound, but in my experience, they are often not sensitive to these.

Problems

Low Pet Engagement →	Dogs were not naturally attracted to the device, leading to low playability and minimal interaction.
High Learning Difficulty →	The button-based interaction required training, which most urban pet owners do not have time for.
Weak Response to Virtual Sounds →	Dogs showed little sensitivity to virtual audio cues, making sound-based interaction ineffective.

What dogs Love ?

I conducted research on the four best-selling and most beloved toys by dogs on the market.

Chew Toys

- Made of durable materials such as rubber, nylon, or rope.
- Satisfy dogs' natural chewing instincts, relieve teething discomfort, and prevent destructive behavior.

Plush Toys

Soft, cuddly toys, often with squeakers. Provide comfort and companionship, especially for puppies and anxious dogs. Easy to damage for aggressive chewers but loved for emotional support.

Puzzle Toys

Require dogs to solve problems or manipulate mechanisms to access hidden treats. Provide mental stimulation, reduce boredom, and improve problem-solving skills.

Interactive Toys

- Designed for play between dogs and their owners (e.g., frisbees, tug ropes, fetch balls).
- Some products incorporate smart technology (automatic ball launchers, remote-controlled toys).

Sensory Appeal

Chewing Instinct

Owner Involvement

Mental Stimulation

Physical Interaction

Inspiration

Electric fishing machine is a classic interactive toy

- **Dynamic opening and closing mechanism**: The fish's mouth constantly opens and closes, creating a momentary "catch/miss" challenge.
- Circularity: The turntable rotates continuously, ensuring continuous play and avoiding monotony.
- Combining fun and challenge: Players must seize opportunities, testing their concentration and reflexes.

Structure and working principle

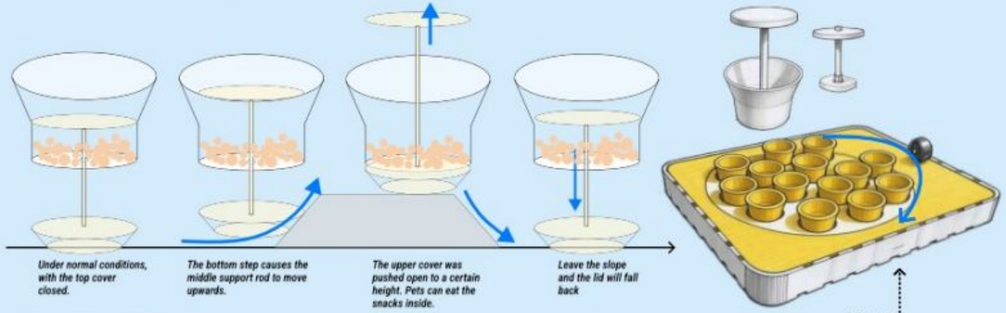
A small motor is built in to drive the disc to rotate.

Multiple fish nests are evenly distributed on the turntable.

The raised track on the base controls the opening and closing of the fish mouth.

It opens and closes by gravity and contains magnets inside.

Structure Design

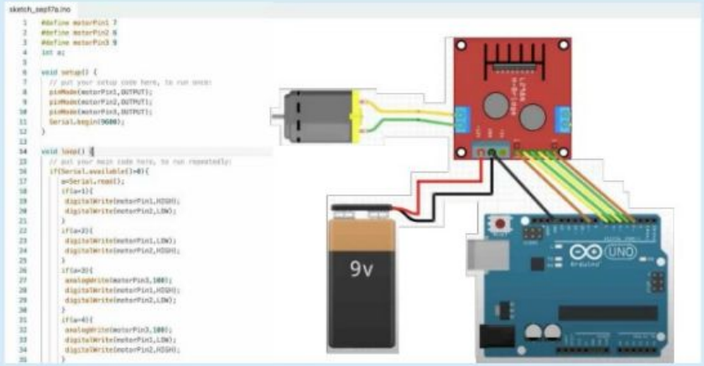


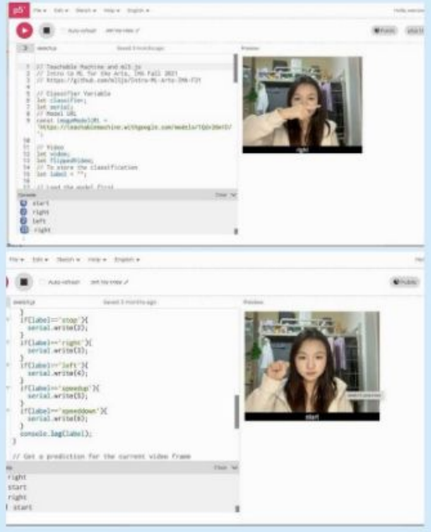
Working Diagram

This toy is driven by a dog's sense of smell. The owner can monitor and control the dog's play through an app.

Making Process

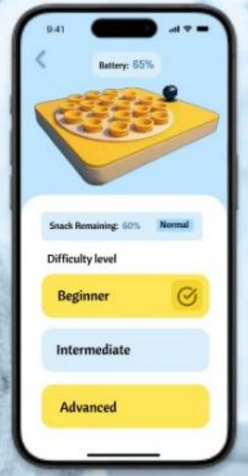
I made an interactive prototype with Arduino, controlled the rotational speed of the motor through L298N, and recognized the basic game gestures with the help of Teachable Machine.
Then, the recognized data was converted into a start signal through P5.js and sent to Arduino, thereby enabling the owner to remotely control the toy.





Interface Design

The owner can remotely control the speed and direction of the toy's rotation through gestures or touch. On the left is the real-time footage captured by the camera.



The automatic mode of the toy and select the difficulty level according to the actual situation of the dog.

The remaining battery power of toys and snacks can all be checked on the app. If it is too low, an automatic reminder will be given.

The camera records the dog's behavior for data analysis, allowing the owner to see the dog's overall health status at any time.

Automatically generate daily health reports for dogs.

The camera records the dog's behavior for data analysis, allowing the owner to see the dog's overall health status at any time.

System will organize the daily shooting materials and automatically generate dog play replals and daily Vlogs.

The owner will never miss any of the dog's shining moments.

Storyboard

1. Fill each small container with snacks

2. Click Start and the turntable starts spinning

3. Each lid is randomly opened

4. The owner watches the dog playing remotely through the camera

5. The owner can control the direction and speed of rotation on the app

6. The camera records the dog's home and play status, generates a daily health report, and a daily vlog

User Feedback & Testing

"I usually work long hours and always **feel guilty about leaving my dog alone** at home. With this device, I can actually **play with him during my break**, not just check a camera. Controlling the turntable with gestures feels surprisingly natural, and I can see my dog getting excited whenever the lids open.

The daily vlog and health report also make me feel more connected to his life. It really reduces my anxiety and makes me feel like I'm still part of his day."

Johanna
Financial analyst

"My puppy is super energetic and easily bored. This toy keeps him mentally engaged **much longer than regular puzzle toys**. The random opening lids keep him curious, and the rotation speed adjustment is great because I can **start slow for training** and make it harder as he grows.

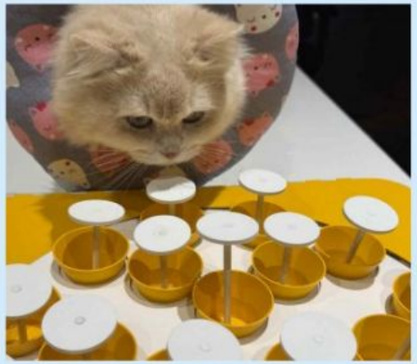
Watching him sniff around and try to catch the treats through the app is adorable. It feels like a co-op game between us even when I'm not home."

Tommy
Game designer

"I've tried many smart pet products, but this one finally feels 'alive.' Most devices only let you feed or watch your dog, but this one actually **lets you interact**. The gesture-control remote play is fun and responsive, and my dog **clearly reacts to the movement** more than to virtual sounds from other devices.

The replay and activity data help me understand his habits better. This is the first device that makes long-distance pet interaction actually meaningful."

Judy
College student



Usage Scenarios

Cecelia is a college student living away from home. When she was at school, she would miss her dog tony very much.
This is the scene of her remotely interacting with tony in the school self-study room using WoofMate.

GREEN COMBO

Board game | Sustainability | Service Design

This board game draws inspiration from mahjong, one of the most familiar and beloved traditional games among Chinese people. By transforming the rules and mechanisms of mahjong into a system that integrates sustainable knowledge, purchasing strategies, and industrial mechanisms, the project aims to make sustainability education more approachable and engaging.

Through a playful and culturally resonant format, the game seeks to help the general public gradually understand the logic and value of the sustainable industry. As sustainability is still in its early developmental stage in China, this design serves as a bridge — promoting public awareness and participation through an accessible, everyday form of play.

CONTEXT

Sustainable Industry in China

Sustainable stores are now on the rise in China, with retail spaces centered around environmental protection, recycling, and local production appearing in more and more cities.

I was attracted by the interesting information so I visited sustainable stores in Shanghai and Guangzhou.

But when I actually visited these stores, I found that the actual situation of the stores was different from my initial expectations.

Very low foot traffic

Even on weekends or in popular commercial districts, the number of customers remained very limited. In some cases, there were even more staff than visitors.

Lack of purchasing atmosphere

Most people only browsed casually and rarely made actual purchases.

Why do these stores, which clearly have appealing designs and values, still fail to attract more people?

Initial problem define

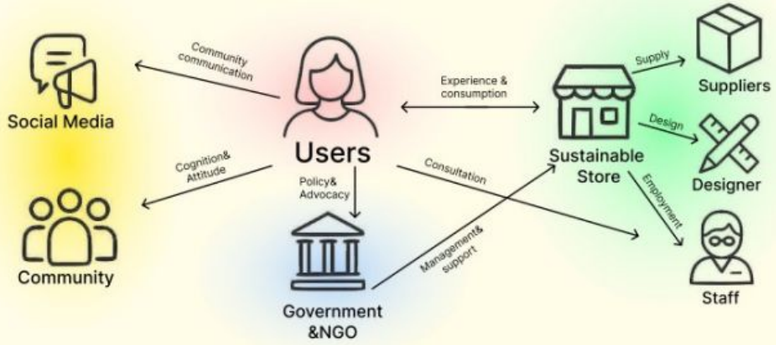
Guess

1. **Site Selection**
   Most sustainable stores are located city centers, while the general public (novice users) rarely pass by.
2. **Price**
   Due to their high costs, eco-friendly products are often 20–50% more expensive than those sold in regular supermarkets.
3. **Shopping experience**
   Most stores are display or sales-oriented, without any interactive elements. Customers cannot intuitively understand "why this is sustainable".

The store's reach is insufficient, and people lack channels for relevant experiences.

IDEATION

I think the key factor in communication and experience lies in the user.
So I first made a Stakeholder Maps around the user group.



FINDING

Through analysis, it can be found that although there are many roles involved, the entire system is still in the initial stage of development in China.

Difficult

The promotion at the government and store levels is still in the process.
Although policy advocacy has emerged, it lacks systematic support.

↓

require a considerable amount of social capital and industry resources to be involved.

↓

For a student designer, the space for direct intervention is relatively limited.

More feasible

Users are not only the core recipients of this system but also the group most likely to be effectively influenced.

↓

As a designer, I can better understand the psychological and behavioral characteristics of users and have a real impact at the cognitive level through design practice.

so User Experience is the most crucial and operational entry point.

USER JOURNREY

To better discuss the user experience, I conduct a user journey to further explore.

Stage	Awareness	Consideration	Store Entry	Product Browsing	Purchase	Post-Purchase Engagement
Touchpoints	- Xiaohongshu, Douyin short videos	- Brand's WeChat/official site - Flyers/posters - WeChat communities - Offline pop-ups, community events	- High-visibility store signage - In-store navigation & guidance - Staff interactions - Product display zones	- Shelves & eco-information cards - Trial zones (touch & try) - Staff consultation	- POS system - Digital receipts - Mobile payment (WeChat/Alipay) - Social media sharing	- WeChat Mini Program membership - Email newsletters, loyalty programs
Needs	- Understand what a "sustainable store" is - Learn about environmental concepts and sustainable living - Compare sustainable vs. non-sustainable products	- Know product categories, value-for-money - Education on sustainability standards - Easy-to-understand signage	- Clear zoning and guidance to quickly find products - Trial and experience-based shopping - Understand sustainability labels	- Easy-to-read product information (eco-labels, ingredient sources) - Quick purchase recommendations (bundles, trending picks)	- Reasonable pricing - Convenient checkout (mobile, membership perks) - Visible sustainability value (loyalty points, traceable impact)	- Motivation to re-engage - Reward mechanisms (loyalty points, eco-credits) - Clear sustainability storytelling (impact visualization)
Emotions	Curiosity, doubt, indifference	Growing interest, intrigue	Excitement, anticipation	Curiosity, delight, possible decision stress	Satisfaction, positive anticipation	Recognition, pride, sharing motivation
Pain Points	- Concept feels abstract - Information overload - Concerns about "greenwashing"	- Products priced higher than regular items - Confusion about eco-standards - Lack of clear value proposition	- Stores feel more like exhibitions than retail - Lack of interactivity	- Product info cards too wordy - Price anxiety - Missing immediate eco-impact feedback	- Single-purchase satisfaction only - Lack of sustained engagement triggers	- Rewards not exciting enough - Sustainability message feels distant

People lack relevant channels to understand sustainable stores which Double check my previous guess

INSIGHT

What causes people to have a cognitive gap about sustainable stores?

- **High user awareness threshold**
  Users lack an intuitive understanding of the value of sustainable stores and struggle to grasp the "why" of going there.
- **Stores have limited communication resources**
  Most are small, independent brands or startups with minimal promotional budgets.
- **Media communications lack content packaging**
  Mainstream media coverage remains focused on consumer trends, fashion, beauty, technology, and other areas.
- **Inadequate education and public outreach.**
  The promotion of "sustainability" in schools and public education is weak, remaining largely at the conceptual level.

REAL VOICE FROM PEOPLE

To figure out people's specific thoughts, I interviewed 25 people, ranging in age from 10 to 70, about their views on sustainable stores in China.

There are 18 people among them **Don't know Fuzzy concept**

Li yanpin
Administrative staff
46

Insights into sustainable product

Something that can be used for a long time
Less pollution to the environment

Attitudes towards sustainable stores

I think these products are good, but when I shop, I still have to consider the price first.
I will not pay simply for the concept.

Have bought sth before?

I'll buy some refills, but my main goal is price-performance, not sustainability.

I had no idea whether these products were sustainable when I bought them

Know but haven't been there *There are 5 people among them*

Bonnie Huo
Art Teacher
23

Insights into sustainable product

Recyclable and reusable
The materials are made from some plants

Attitudes towards sustainable stores

I know it. I will go check it out when I have time.

Have bought sth before?

I have bought some, such as handmade soap. But I didn't think it was very useful, so I didn't buy it again.

I feel like there's too little publicity.

There are 2 people among them **Know and have been there**

Roxy Huang
Product manager
32

Insights into sustainable product

Sustainable products refer to those that have a relatively small impact on the environment during production, use and disposal, and are in line with the concept of sustainable development.

Attitudes towards sustainable stores

The ideal sustainable store needs to demonstrate consistent "sustainability" in every aspect that consumers can experience, such as product selection, store design, and event planning.

Have bought sth before?

I bought a bag made from discarded tires at a sustainable store in Shanghai. It's very special and I like it very much. I also bought a soap made from herbal plants in Guangzhou.

A warm and pleasant community atmosphere can better attract environmental protection enthusiasts to join.

FINDING

1 **72%** of people are unaware of sustainable stores.

2 Especially the **middle-aged** and **elderly**, have a rather vague understanding of the concept of sustainability.

→ What people really lack is **education** about sustainability

DECISION MAKING

The four stages of sustainable store growth

INSIGHT

If no one knows or understands the significance of a "sustainable " early on, subsequent promotion, expansion, and community building are impossible.

Challenge

- **Stores have limited funds and manpower, making it difficult to implement large-scale traditional advertising.** → **How can we transform knowledge into a fun, memorable, and reusable experience at a low cost?**
- **Social media topics struggle to expand beyond a niche audience, lacking easily replicable communication channels.** → **How can we ensure that communication has a "social amplification effect" (users are willing to influence others)?**
- **Current education systems are often rigid and lack practical relevance, making them difficult to connect with the public.** → **How can we transform knowledge into a fun, memorable, and reusable experience at a low cost?**

Design proposal

HOW MIGHT WE	**Enhance people's understanding of sustainability in a low-cost and easily acceptable way**
FOR	**People who do not understand the concept of sustainability and stores**
SO THAT	**Help people understand sustainable living and be willing to enter sustainable stores**

DESIGN INSPIRATION

Easy to accept and low cost, these key words remind me of the traditional Chinese game

Mahjong

Popularity & Accessibility

Mahjong is extremely popular in China and East Asia. It's not just a form of entertainment, but a cultural symbol. Mahjong is often played at family gatherings, community events, and friends' dinners, fostering a cross-generational connection.

Physical Attributes

Through concise and eye-catching fonts, symbols, colors, and layout, users can quickly understand abstract rules and cultural logic during use.

Interactive & Strategic Gameplay

The joy of mahjong lies in the interplay of luck and strategy. Players collect, combine, and exchange tiles through the "pong, kong, and hu" method.

EXPECTED IMAPCT

Sustainable store + Mahjong = Sustainable educational board games

While playing board games in a relaxing and entertaining way, users gradually understand "what a sustainable store is" and are encouraged to go into sustainable stores and make purchases through a reward mechanism.

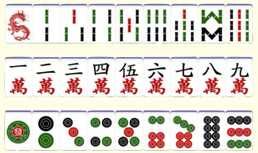
PROTOTYPE

Card Design

Knowledge that sustainable stores need to popularize

What is sustainable → Action card

What's in the Sustainable Store → Commodity card



Each type of card corresponds to a suit of mahjong

Rule Design

Arrange the cards in your hand into a winning formation as quickly as possible

At the start of the game, each player draws 7 cards. Then, in turn, each player draws one card and discards one card during their turn. The player wins when their hand forms a winning hand after drawing a new card.

HOW TO WIN? HU!

Two sets of different types of Commodity cards (three each)+ two Green action cards

Same categories commodity cards 3 | Same categories commodity cards 3 | Green action cards 2

Strategy Pong !

When another player plays a card that matches two cards in your hand, you call "Pung," and then place those three identical cards on the table to form a triplet.

Attention

- Players need to determine for themselves whether the behavior described on the behavior card is a green behavior.
- The first person win will receive a coupon for spending at a sustainable store.

TEST & FEEDBACK

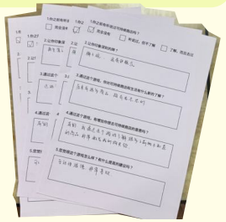
Group 1

Test group: Young people. They all have experience in playing mahjong.

There is a behavior of using disposable tableware when ordering takeout. I often do this.
In my mind, it is not environmentally friendly, but now I have gradually forgotten it in my life. This game has made me realize again that it is a bad behavior.

INTERVIEW

People were particularly impressed by the "Coconut Bowl" and "Organic Vegetable" cards; their names were short and the designs were interesting.



QUESTIONAIRES

Group 2

Test group: Family. There was a little kid who had no experience with mahjong.

IMPROVEMENT

- **Add a product description next to the card so players can read it while waiting.**
- **Increase the number of cards of the same type for each category. The situation where the cards on the field run out before anyone wins.**
- **The categories aren't clearly distinguished, customize product categories according to different stores, centering on people's lives.**
- **Actions could be added that might go unnoticed in daily lifebut are actually not environmentally friendly.**

MAKING PROCESS

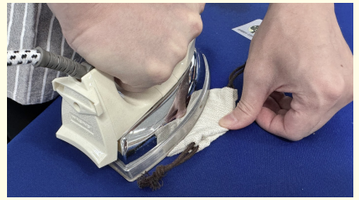
Material Selection

Reuse waste paper to make it

Tear up the waste paper and put it in a basin. | Add an appropriate amount of water. | Put it into the blender to form pulp. | Filter the pulp.

Use a mold to lift up the pulp. | Use a sponge to absorb the excess water. | Dry it in a cool place. | The recycled paper is done.

Package Design



Card Design

I tested Green Combo in the nearby community. The elderly and children often play here.

I pasted the poster on the information wall, so almost everyone who comes here can see it.

Use the discount coupon to make purchases.

"The card designs are quite nice, how to play it?"

She was the first person drawn in by the poster and received the Green Combo board game.

Wait a moment! I can Pong! I have two tyre bags!

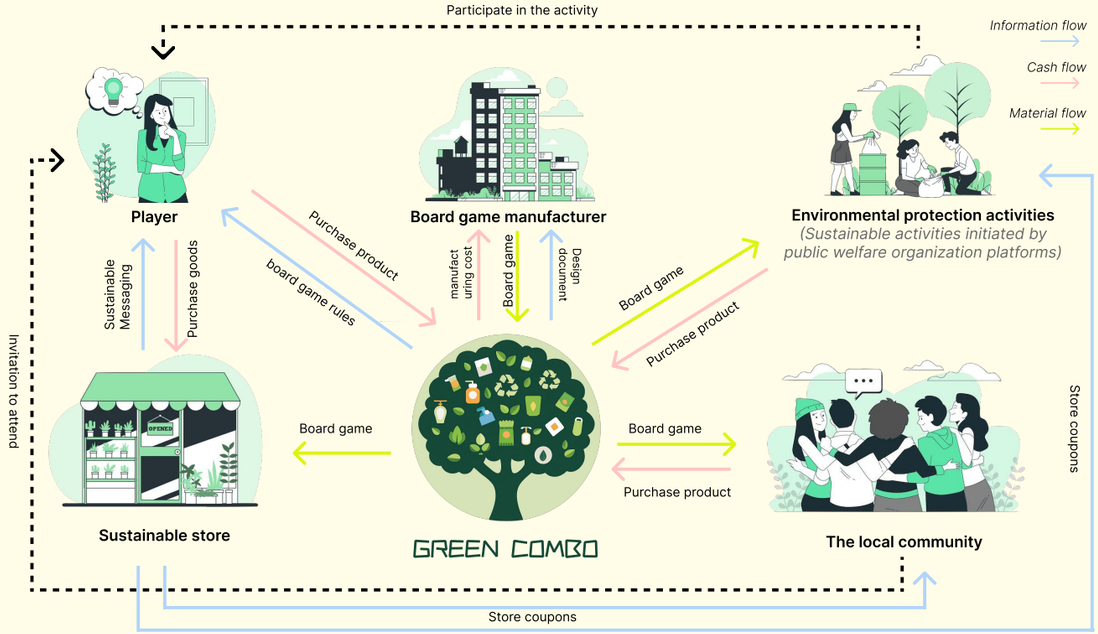
Why are you so lucky today?

The children were also attracted by these cards and were carefully looking at the content of each card.

Green Combo is very portable You can hang it anywhere!

PRODUCT DISPLAY

SYSTEM MAP



Future Planning & Development

Step 1

Testing & Optimization

•Conduct multiple rounds of playtesting with diverse age groups (families, students, communities).
•Gather feedback on rules, difficulty, and educational impact.
•Refine and release an optimized version.

Step 2

Online Promotion & Partnerships

•Promote through social media (Instagram, TikTok, Xiaohongshu, Weibo).
•Collaborate with sustainable shops to design exclusive product cards representing their goods.
•Increase visibility and customer engagement for partner shops.

Step 3

Education & Expansion

•Introduce the game to schools and community events as a sustainability education tool.
•Develop a digital version (mobile app or mini-game) for broader reach.
•Partner with NGOs, creative brands, and board game cafés to host “Green Combo Challenge” tournaments.

STORYBOARD

1. The local residents were attracted by the promotional posters and became interested.

2. The residents got the board game cards and planned to try them with their friends.

3. Everyone plays the game together, and the winner can get a discount coupon for the sustainable store.

4. Players come to the sustainable store according to the address on the discount coupon.

5. Players select their favorite product in the store.

6. Players use discount coupons to settle the bill and complete a purchase in the sustainable store.

USER FEEDBACK

Operators of sustainable stores

Age 30

Q1: Do you think this game is helpful for sustainable stores to attract customers?

This is a great idea that combines the products and concepts of sustainable stores with board games. Let everyone gradually get to know our store while playing. I would be willing to cooperate with you if possible.

Q2: Do you have any suggestions for improvement?

Perhaps we could create an online version similar to a mobile game, so people can play it directly on their phones.

Q1: What do you think of this game? Have you learned anything about it?

I think the words under that card are a bit small and I can't see them clearly. Maybe it's because I have presbyopia. This game like Majiang, but a little simpler. I saw organic vegetables and unpackaged grains here, all of which were very attractive to me.

Age 66

Q2: Do you think this kind of game is appealing to older people?

Yes, especially to those of us who enjoy playing mahjong. It's easy to learn and we can even get coupons to buy things.

Retired elderly

Primary school student

Age 12

Q1: Do you think this game is fun?

Yes, it's fun! It's like collecting cards, and you win by assembling a 3+3+2 combination. But I was really unlucky last time; I didn't draw any clothing cards, otherwise I would have been number one!

Q2: What did you learn from the game?

I learned what product are sold in the sustainable store, and some behaviors like cycling are environmentally friendly and minimizing the use of disposable products. I want to buy that coconut shell bowl; it looks so cool!

MEMOJI

MEMOJI is an AI-powered family communication service app for seniors. Through voice emotion recognition, personalized emotional expression generation, and a private family social circle, it helps seniors rebuild a communication experience of "being seen, understood, and responded to" in the digital age.

The project addresses the digital social gaps faced by many senior citizens in the digital age:
1. They habitually use voice to express themselves, but are often ignored.
2. They crave cross-regional emotional connections, but can only send "simple text messages."
3. They fear new technologies, yet fear "falling behind the times."
4. They worry about online scams, leading to increasingly withdrawn social interactions.

MEMOJI is not just an emoji tool, but a warm gateway for seniors to enter the AI era.

BACKGROUND

In the process of getting along with the elders at home and observing their wechat usage habits, I have repeatedly witnessed the following phenomena:

Often send voice messages, but family members don't always listen.

"I send a voice message to my granddaughter, but she forgets to open it when she's busy." The elderly are used to expressing their emotions through voice messages.

11"

Very different from the emojis that young people commonly use

They prefer sending emojis with blessings, holiday themes, and flower motifs. They don't understand the silly memes used by younger people in family group chats.

Afraid to add new friends deeply affected by anxiety about scams

The lack of identity verification mechanisms leads to fear of being scammed, resulting in reduced social interaction and ultimately a "digital defense state."

I'm curious about AI but lack confidence

"Nowadays everyone is playing with AI. I also want to be able to use it, but it's too difficult."

When young people talk about AI, I can't even join in.

AI

Observation Insights

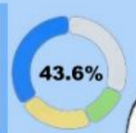
1. **Older adults lack ways to express themselves without requiring the other person to spend time seeing their message.**
2. **Older adults have a weak sense of social security, causing them to withdraw from open social interactions.**
3. **There is a natural gap between older adults and younger people in their digital communication systems.**
4. **Older adults strongly desire to "keep up with the times" by learning new technologies, but lack confidence and the tools to adapt.**

RESEARCH

Intergenerational communication difficulties:

of elderly respondents in Hong Kong were categorized as "socially isolated," with an average social network score of only **7.6 out of 30**

of the elderly were afraid of "being seriously ill and having no one to care for them," followed by "having an accident and having no one to help them."



The Loneliness Black Hole

68.3% of Hong Kong's elderly feel lonely, significantly higher than the global average of **34%**. Hong Kong's highest-ever single case of telecom fraud resulted in losses of **HK$250 million**.

Older adults face the dual pressures of loneliness and security risks in digital social interactions.

CASE STUDY

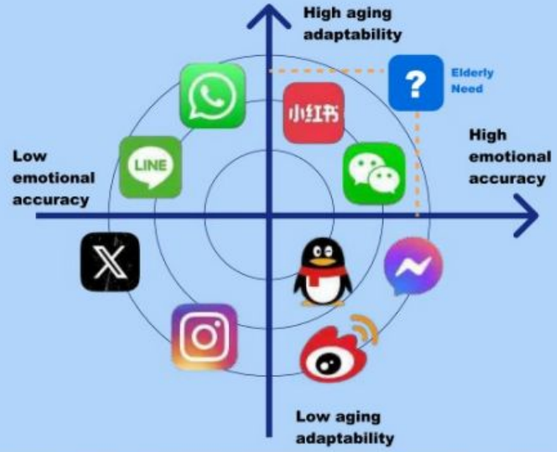
Telecom Fraud Threat

January 2024
In Sichuan Province, China

19 elderly people were victimized (most of whom lived alone or whose children worked away from home), involving 1.992 million yuan.

Social safety for older adults requires "closed, trustworthy networks of relationships," not more tools.

MARKET & PRODUCT ANALYSIS



The mainstream social tools are designed for universal use and lack the "exclusive emotional expression ability for the elderly".

Insight

This analysis reveals a clear unmet need

Older adults need a tool that combines:

- **High emotional accuracy**
- **High aging adaptability**
- **Low learning cost**
- **High safety & trust**
- **Intergenerational communication support**

→ **A unique category that does not exist in mainstream social platforms today.**

PERSONA

Based on the above research results, we interviewed 10 elderly people aged 60 to 75 to better understand their true demands. I summarized and classified them into three persona.

Name: Mei Wang
Age: 68

Living situation: Lives alone; children work in other cities
Numerical ability: Familiar with spoken language, but types slowly

"I send voice messages to my daughter, but she might not want to listen, and sometimes she'll forget to listen when she's busy. When she doesn't reply, I wonder—am I being too verbose? Am I bothering her?"

Expressing myself isn't the problem; What I worry about most is being seen and heard. If my family can't see my voice messages, I feel like "what I say doesn't matter."

Name: Liang Chen
Age: 65

Living situation: Lives alone; daughter works overseas
Numerical ability: Type slowly, prefers sending photos

"If I could see my daughter directly, that would be best... so she can feel me, not just see the words." "When I type something, it feels cold. Words don't carry the warmth I want to send her. I want her to feel that I miss her, but typing doesn't help."

I don't struggle with conversation—I struggle with the lack of emotional realism in digital communication.

Name: Yushan Lin
Age: 70

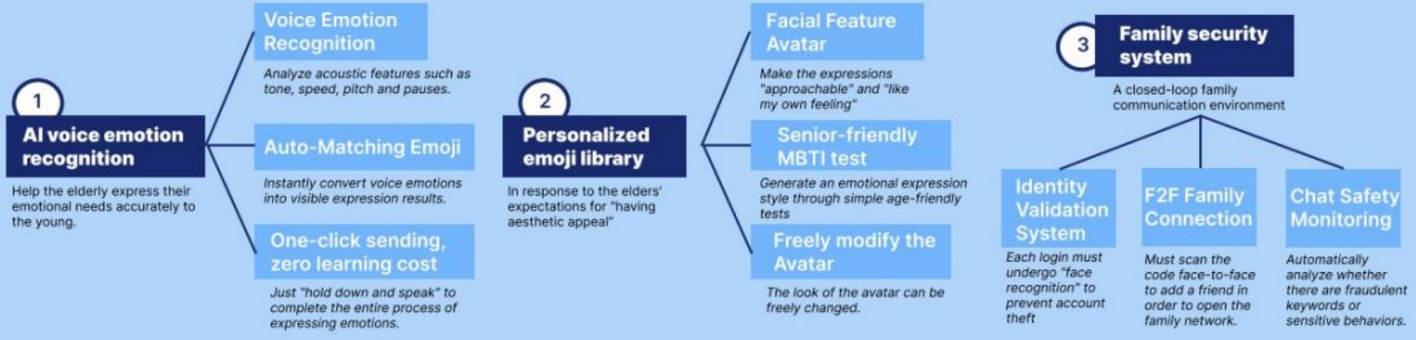
Living situation: Lives with her grandchildren
Numerical ability: Enjoys photos and music; not familiar with emoji culture

"The expression I imagine isn't just moving—it should have the music I like. My mood needs music." "Those emojis young people use... I don't really understand them. I want something that feels more like me, that can show my feeling, not just a generic face."

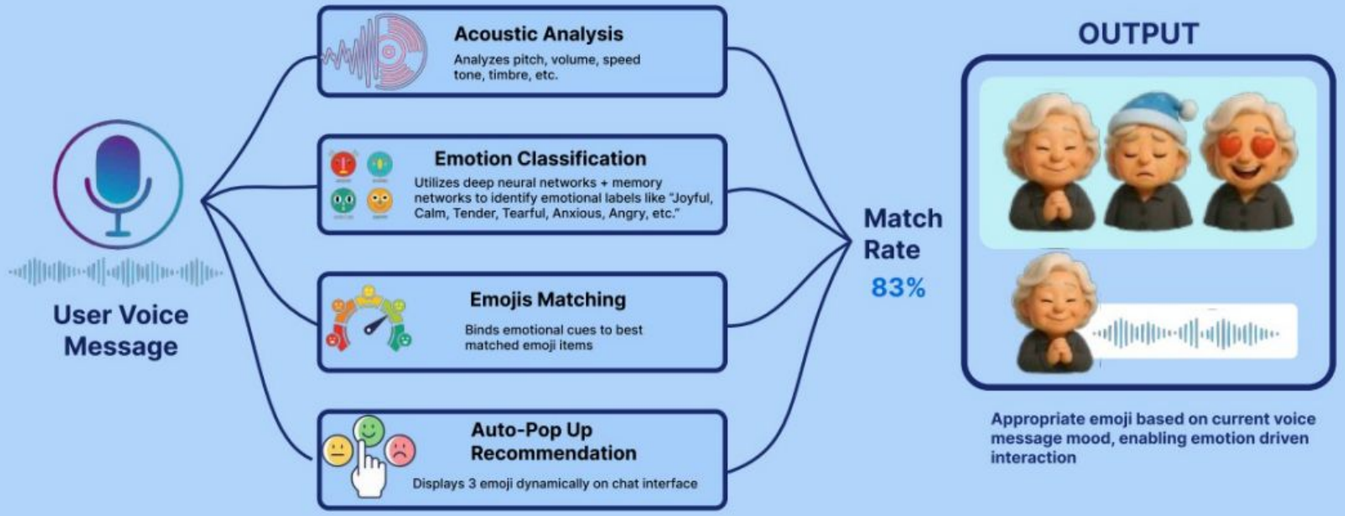
I don't lack tools—I lack tools that can reflect my personality.
My emotional world is richer than what standard emojis can express.

DESIGN DIRECTION

Based on the research results, I have summarized the three major design directions required for the app and the corresponding design strategies.



GENERATIVE LOGIC

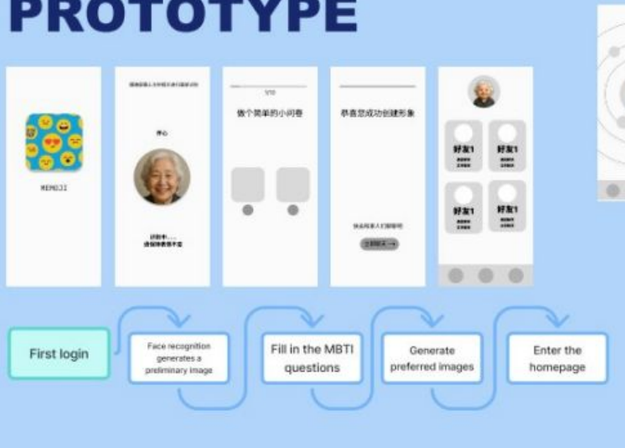


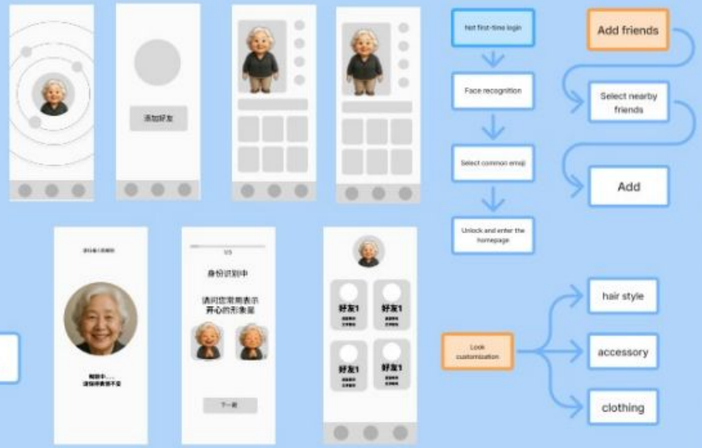
Appropriate emoji based on current voice message mood, enabling emotion driven interaction

USER JOURNEY

	INITIAL USE	CONTINUED USE		DEEPENED USE
User Goal	Learn to use emojis	Get attention	Feel comfortable in safe social	An active participant in family chats
User Behavior	"This little emoji character is quite cute... let me try sending it to my granddaughter to see if she likes it."	Will this meme be seen and stolen by a stranger?	There is very little junk information and it is not annoying to use.	"I'm the main character of our family chat group now!"
User Emotion				
Memoji support	Input: Face detection and emotion classification model after a user uploads a single personal photo.	Encryption: Equipped with acoustic feature extraction functions such as pitch, timbre, and volume to assist in encryption.	Emoji matching engine: Bind emotion tags to the best match in the emoji library.	Automatic pop-up recommendation: The chat interface displays the corresponding emoticons in real time.

PROTOTYPE





MBTI test questions

Successfully created your own virtual emoji

Home Page
Display the contact person and chat method

"Add Friend page
Only your family and distance will be seen

Personalize your personal emoji

Voice chat interface

Voice-to-text interface

STORYBOARD

USER FEEDBACK

Elderly Users

Voice-to-Emoji AI Feature — "Very helpful"

"I like that I don't need to type.
When I speak, it shows a little face with my mood.
That saves me time and makes me feel understood."

Personalized Emoji — "Feels like me"

"The emoji looks like me! It makes me want to send more.
It warns me when something looks strange.
I don't dare add people on WeChat, but here I feel safer."

Younger Family Members

Emotion Visibility — "I can see how my mom feels instantly"

"Instead of long voice messages, I get a sticker that reflects her mood.
Even when I'm busy, I know she's okay."

Automatic Sticker Suggestions — "Surprisingly accurate"

"The recommended emoji actually fits her tone.
It makes our chats lighter and more fun."

WHAT'S NEXT ?

Petified™ Market

Installation design | Visual Design | Material Experiment

More and more people are beginning to view non-traditional objects like toothpaste, rocks, and cardboard boxes as companions. This absurd relationship ironically challenges the logic of "everything can be commoditized." However, the current consumer market is gradually emerging with products and services catering to non-traditional pets.

Inspired by this, this project creates an "alternative pet store." Through a custom cash register device, product packaging dissolves, revealing its true form and the emotional value attached to it. This installation visually exposes the logic of shopping, prompting viewers to reflect on the manipulation of emotions by consumption.

https://youtu.be/fI4kmfCkGuk

Research

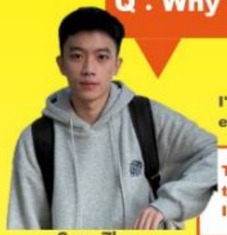
The definition of "pet" among Gen Z is becoming increasingly diverse, including practices such as "cloud pet ownership," virtual pets, and even treating plants, plush toys, or AI robots as pets.

Social media sharing posts

The "pets" community on TikTok/Xiaohongshu is spreading content about people using inanimate objects (rocks, cardboard boxes, yeast) as pets.

Interview

Q : Why do you want to get a non-traditional pet?



Sam Zhu
Age: 28
Pet: toothpaste

I'm exhausted by the time I get home every day, and I don't have the energy for a cat or dog. But I still need a companion.

Toothpaste doesn't make noises or die. Every day while I brush my teeth, I talk to it, and it actually feels like it's protecting me. Ultimately, I might not be raising a pet so much as cultivating a consistent ritual.

Lily Wang
Age: 25
Pet: Stone

Look, the pet market is full of scams these days. A 'smart feeder' isn't selling functionality, it's selling emotional manipulation. Why would I spend thousands of dollars on something for 'emotional comfort'? I won't! I keep a rock. It has no packaging, no marketing, just itself.

The meaning of the rock being with me is that it's nothing. It's a way of refusing to be manipulated by consumerism.

Conclusion

- **Rebelling against** emotional consumerism → Refusing to be **exploited** in absurd ways
- Feeling of **powerlessness** in a repressive society → Filling **companionship** with non-traditional pets

Market Research

The current consumer market has gradually witnessed the emergence of products and services catering to non-traditional pets.

On Etsy

Dozens to Hundreds

of commodities listings in the "pet rock" category

On Taobao and Xiaohongshu

There are also a large number of no-traditional pets for sale.

Case Study

The "Rock Pet" craze

In 1975

Gary Dahl

Each pet stone comes with a cardboard box with air holes so that the "pet stone" can "catch its breath".

An instruction manual inside, introducing how to take care of the pet stone.

1.5 million

In just six months, Gary sold 1.5 million pet stones and made 6 million US dollars.

Contradiction

The paradox of Emotional Economy

Current market trends have repackaged this resistance into a new form of consumption. Non-traditional pet companionship has now become a "sellable service."

The original motivation for people to own such non-traditional items was a rebellion against sentimental consumerism.

Even rebellion and satire can be absorbed and resold by the logic of the market.

While people express their real emotional relationships through non-traditional pets, they are also drawn into another layer of "sellable emotional game" through algorithmic push notifications on social platforms.

Design Concept

Ideation

I wanted to create a "non-traditional pet store" and design a series of non-traditional pet products, derivatives, and packaging, allowing visitors to "fall" into the trap of consumerism through an entertaining experience.

Ultimately, visitors will need to complete the purchase process through a “cash register device”: they place their selected products into the device, and the packaging will gradually dissolve, revealing the product's true form. At the same time, the device reveals the additional "emotional value" attached to the product.

Service Blueprint

Step	Physical Evidence	Customer Actions	Frontstage	Backstage
Enter the Store		Walk into aesthetic store; attracted by atmosphere	Staff greet; guide visitor	Light, music, scent
Select Product		Select emotional product (mango core, toothpaste, box dog)	Emotional taglines and packaging.	Narrative design, pricing strategy
Checkout		Place commodity into special cashier	Machine open; system scans	Sensor and mechanism activate.
Packaging Melts away		Watch packaging dissolve and peel away	The dissolved liquid flows out	Program detects the dissolution situation
True Form Revealed		See real object—simple and raw	Machine opens to show item.	Mechanic trigger release
Receipt Prints		Receive printed emotional receipt	Printer outputs message	Algorithm generates text
Shock and Reflection		Hold item and ticket; emotional realization	Lights fade; music slows.	Adjust lighting and music

Sketch

Design inspiration:
Cash register toy + vending machine

Cash register toy

Traditional cash register, but the drawer is too narrow to accommodate the packaging melt function.

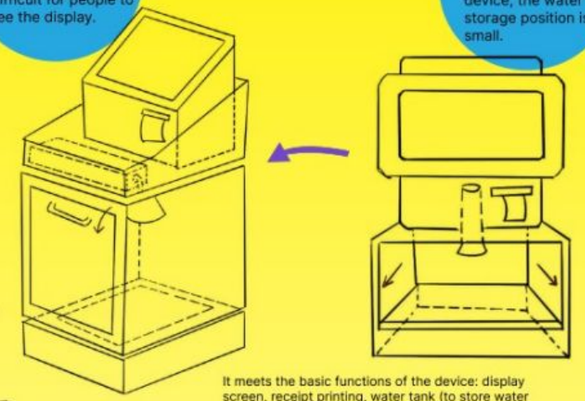
Vending machine

There is ample space to accommodate the melting function of the package.

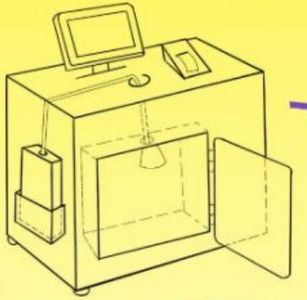
The transparent glass allows for a more visual display of the package dissolving process.

Sketch 2.0
The water tank is larger than the first version.
Difficult for people to see the display.

Sketch 1.0
Mainly referring to the cash register device, the water storage position is small.

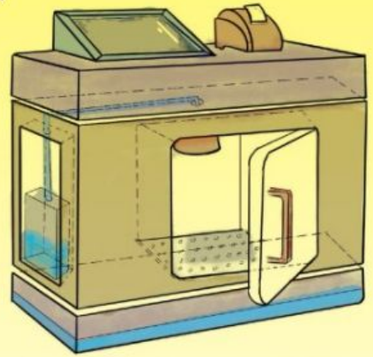


It meets the basic functions of the device: display screen, receipt printing, water tank (to store water needed for dissolution), water pump, and nozzle.



Sketch 3.0
The overall orientation of the device was adjusted, and the water tank was placed outside (to store more water).

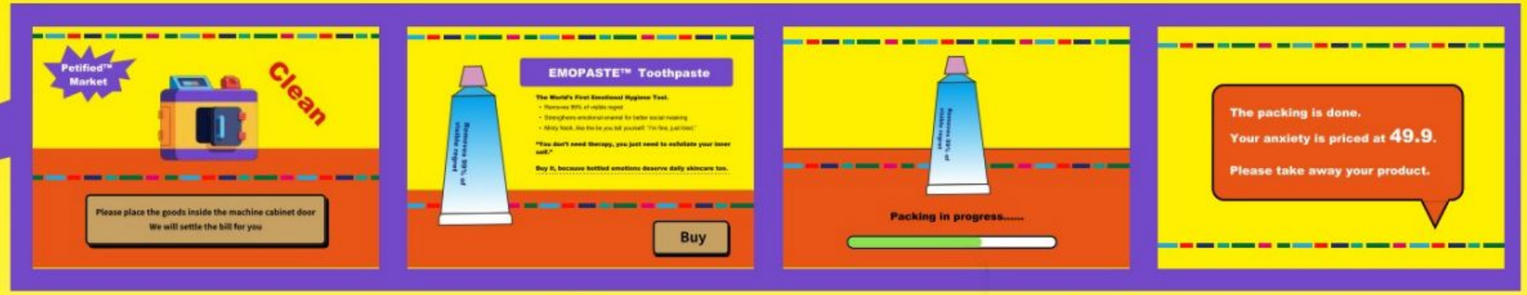
Sketch 4.0



The overall aesthetics of the device is optimized, with the water tank and water pipes built in at an appropriate height to meet the needs of the device.

Installation Design

This installation looks like a cash register on the surface, but when customers put the purchased goods into it, the device will spray water to melt the product packaging.



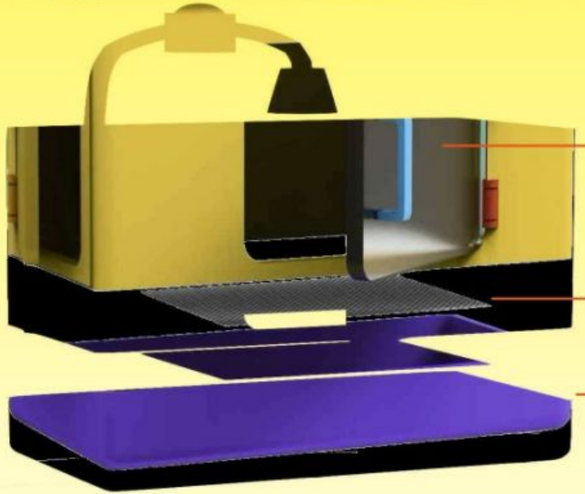
Display
Providing customers with operation tips

Receipt Printer
Print receipt after dissolution is completed

Upper cover

Water pump
Pump water out of the storage tank

Sprinklers
Expand the spraying area

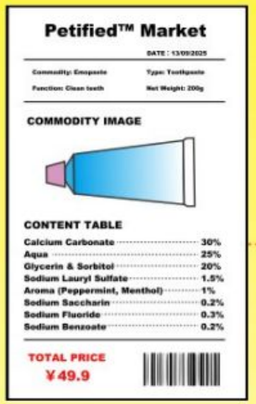


Water tank
store water can dissolve 30 packages

Transparent door
Easy to observe the packaging dissolution process

filter plate
Filter water and dissolve residue

Wastewater storage
Store water and recycle it



Petified™ Market

DATE : 13/09/2025

Commodity: Emopaste — Type: Toothpaste

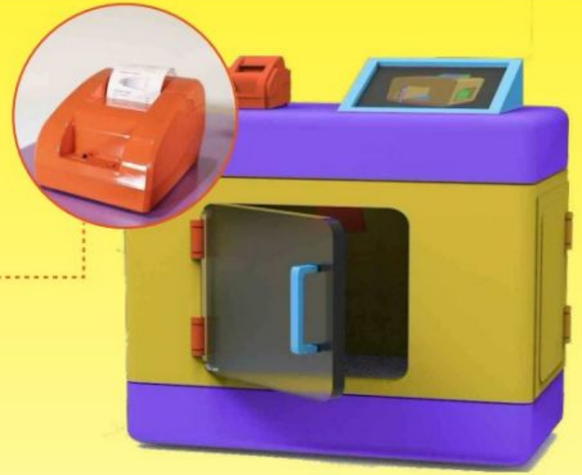
Function: Clean teeth — Net Weight: 200g

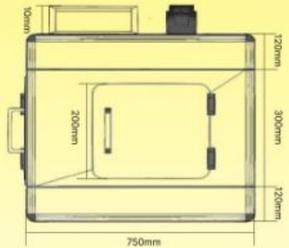
COMMODITY IMAGE

CONTENT TABLE

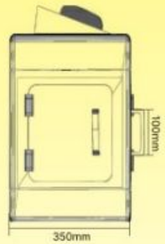
Calcium Carbonate	30%
Aqua	25%
Glycerin & Sorbitol	20%
Sodium Lauryl Sulfate	1.5%
Aroma (Peppermint, Menthol)	1%
Sodium Saccharin	0.2%
Sodium Fluoride	0.3%
Sodium Benzoate	0.2%

TOTAL PRICE
¥49.9

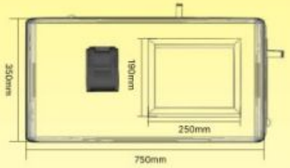




Front view



Side view



Top view

Making Process

Make a product model with acrylic to determine the size.

Use spray paint to paint the model.

3D print some parts and paint them for coloring.

Final Installation

Packaging Design

Ideation

To get satisfactory packaging dissolving results, I tried several different packaging materials.

Hardness: ★ ★ ★ ★
Printing Effect: ★ ★ ★ ★
Water-Solubility: ★

Modified fiber paper

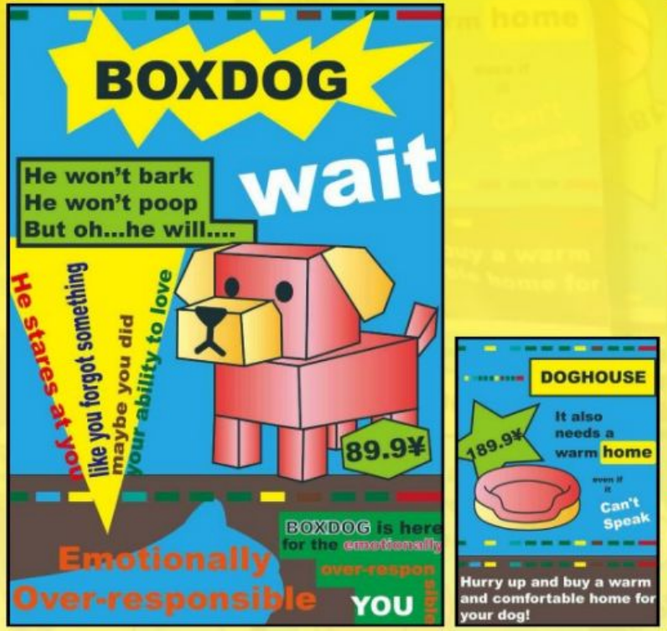
Hardness: ★
Printing Effect: ★ ★
Water-Solubility: ★ ★ ★ ★ ★

Edible rice paper

The most suitable !

Hardness: ★ ★ ★
Printing Effect: ★ ★ ★ ★
Water-Solubility: ★ ★ ★ ★

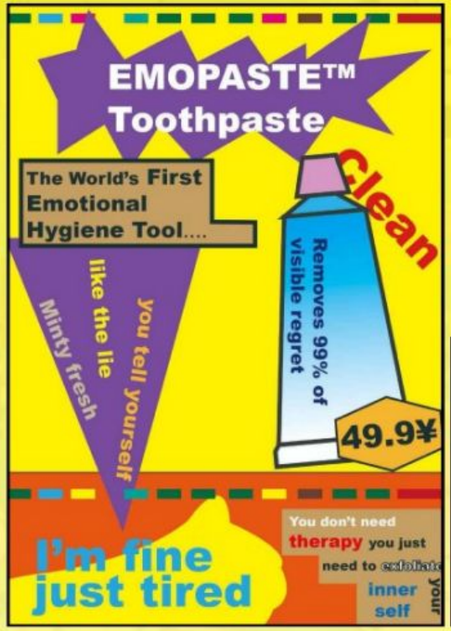
PVA
(polyvinyl alcohol water-soluble film)



BOXDOG & DOGHOUSE

He doesn't need food, walks, or your time.
But somehow... he's always there for you.
Watching. Waiting. Existing only to make your loneliness look elegant.

Always accompany you unconditionally



TOOTHPASTE & CARE GUIDE

"Cleans teeth. Scrubs shame."
Every day you brush your teeth.
But when was the last time you brushed off the guilt of oversharing, the shame of liking his story, or the exhaustion of smiling through the pain?

Bottled emotions deserve daily clean

MANGO CORE & DECORATIONS

"The hard, bitter friend who never leaves."
He doesn't text back. He doesn't need walks.
He just sits there, absorbing your sadness like a seasoned therapist who charges in silence.

Your pain deserves a shape

Exhibition

Video Link： https://youtu.be/fl4kmfCkGuk

When a product is wrapped in layers of allure are we truly buying the thing itself ?
Or merely consuming the emotions and desires that were manufactured for us?

1 Customers enter the Petified market.

2 Customers are attracted by the packaging and selling points of the products and choose them.

3 The customer places the items they decide to purchase into the cash register for checkout.

4 The device recognizes the product and "melts" it, gradually dissolving the packaging.

5 The customer takes out the product with the packaging completely dissolved.

6 The receipt machine prints out a shopping receipt that shows the original appearance of the item.

Is Your Child on the Path to Success?

Life Mancer

LifeMancer is a **critical speculative design project** that satirizes parents' growing obsession with control and the narrow, standardized definition of **"success."**

In a dystopian near-future, brain-computer interface technology lets parents monitor and manipulate children's emotions, focus, and behavior to fit a pre-set success template.

The project questions **who decides a child's future, whether success can be engineered,** and where to draw the ethical line in tech-driven education.

https://youtu.be/Kv3FqHJPKK0

BACKGROUND

Tiger Parenting Intensifies Across East Asia

In East Asia, **tiger parenting—strict, results-driven child-rearing**—is intensifying.

What it looks like ——— Education Competion

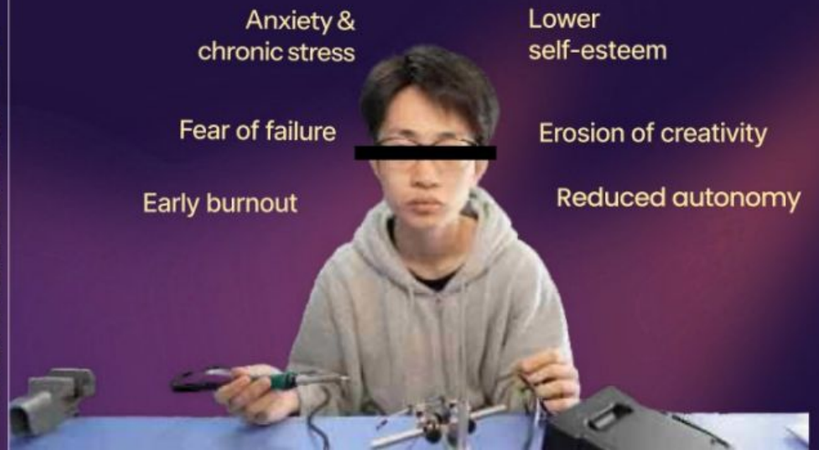
Over the past 15 years, education spending has increased by approximately 185.7%.

RESEARCH

1.Why East Asia Is Fertile Ground for Tiger Parenting

Confucian values, exam culture, and fierce competition make education the prime path to success. while modern uncertainty turns schooling into the only controllable investment in securing a better future.

2.Negative Impact on Children



3.In a race for success, education has become industry.

From guaranteed university admission to assured job placement, **a full-service education market thrives on parental ambition.**

Core of educational institutions

Make you believe：

The rigid definition of success: "good school" and "high-paying job" are the only goals.

4.Possible Future Direction

If things keep going like this, **education may turn into a service that controls every part of a child's life** — from birth to getting a job. In that future, education and control could become the same thing.

Who defines success?

Do children have free will in this world?

Imagine the possible future

In a hyper-competitive future, **parents treat children as property.** LifeMancer's brain–computer chip **monitors and controls behavior** in real time, suppressing distraction, emotions, and noise to enforce a single "success template."

Design Purpose

Satirize extreme parental control and spark reflection on education, values, and freedom.

Core Functions

1. Noise Isolation
Blocks environmental sound.

2. Emotional Control
Releases calming substances.

3.Addiction Prevention
Induces aversion to "undesirable" acts.

CONCEPT---A future educational company

Brand Logo

LifeMancer

Brief

LifeMancer company is committed to providing parents with "life design" services for their children.

Through scientific allocation of education, career planning and life experience, we can fully **meet parents' expectations for their children's "success"!**

How it defines success ?

"The brand defines success by **deconstructing the journeys of accomplished individuals**—where they studied, the positions they held—**turning success into a replicable template.**"

Making Process

Inspiration

Reference to some concept products with a futuristic sense of metal, features: low-key, calm, with a sense of line.

Sketch

Sketch headphone shapes.

Model Making

2D to 3D

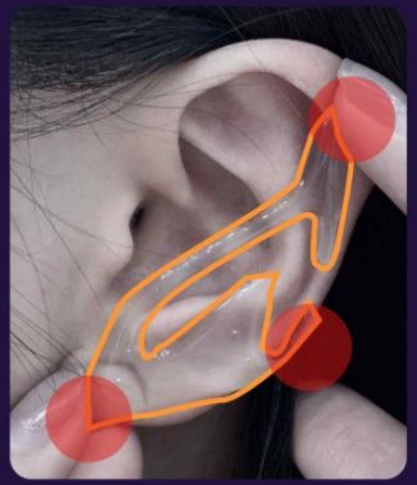
Use uv glue to make simple models

Debugging uv glue

Illumination solidification

User Testing

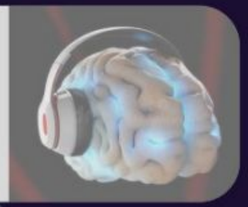
Adjust the size to fit ears



Final Presentation

3D Printing

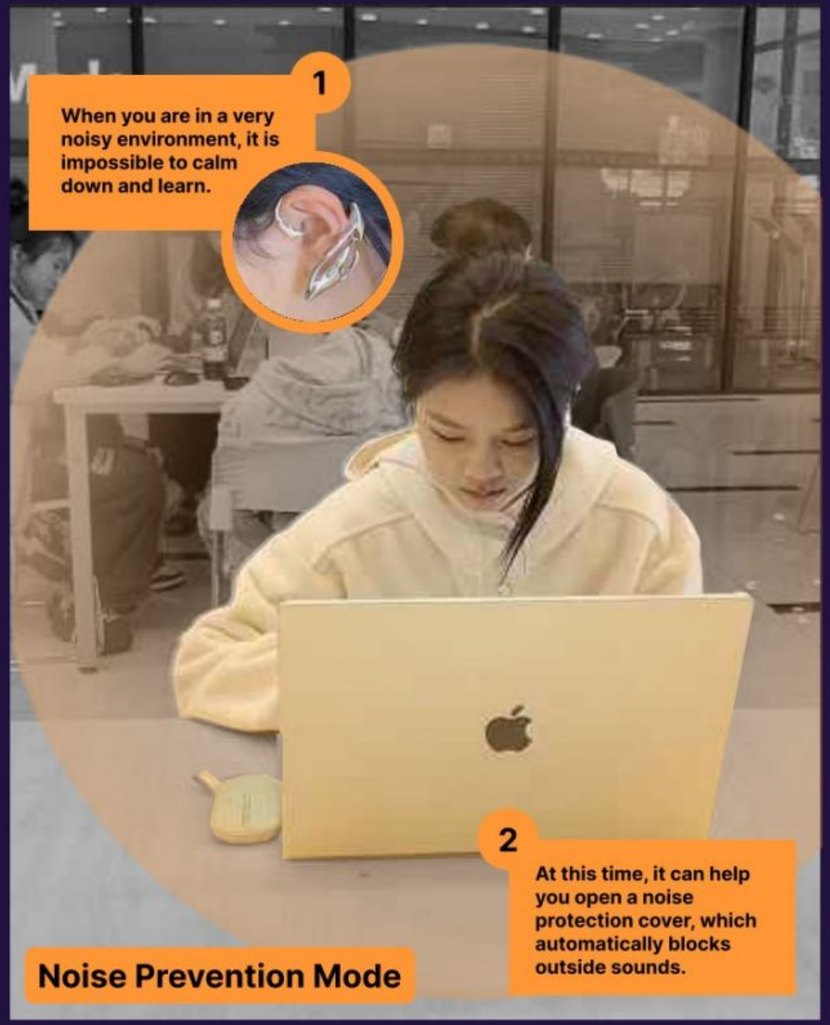
Noise Isolation
Control of auditory nerves - unable to hear any external noise



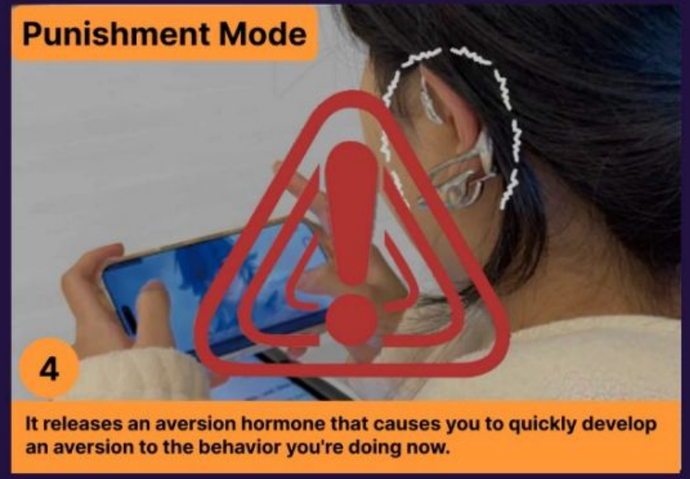
Emotional Control
Brain tranquilizers - quick elimination of bad moods

Addiction Prevention
Release aversion hormones - Every time a bad behavior occurs, it is released immediately.

Scenario


1
When you are in a very noisy environment, it is impossible to calm down and learn.
2
At this time, it can help you open a noise protection cover, which automatically blocks outside sounds.
Noise Prevention Mode

3
When it detects that you're engaging in behavior that interferes with learning.
Addiction Prevention Mode


Punishment Mode
4
It releases an aversion hormone that causes you to quickly develop an aversion to the behavior you're doing now.

Emotional Control Model
5
When it detects that you're emotionally unstable and affecting your studies.

6
It releases a calming substance into your brain that helps you stabilize your mood quickly.